

DAY 11 - SUMMARY

A constructor is a special method in a class that is automatically called when an object is created. Its main purpose is to initialize the object's data.

Key points:

- It has the same name as the class.
- Runs automatically when an object is created.
- Used to set initial values for object properties.
- Does not have a return type.

Types of Constructors:

1. Default Constructor

A constructor with no parameters.

Example:

```
class Student {
    Student() {
        System.out.println("Student object created");
    }
}

public class Main {
    public static void main(String[] args) {
        Student s1 = new Student();
    }
}
```

Output:

Student object created

2. Parameterized Constructor

A constructor that accepts parameters to initialize values.

Example:

```
class Student {
    String name;

    Student(String n) {
        name = n;
    }
}
```

```
        void display() {  
            System.out.println(name);  
        }  
    }  
  
    public class Main {  
        public static void main(String[] args) {  
            Student s1 = new Student("John");  
            s1.display();  
        }  
    }  
}
```

Output:

John

3. Copy Constructor

A constructor that creates an object by copying another object's values.

Example:

```
class Student {  
    String name;  
  
    Student(String n) {  
        name = n;  
    }  
  
    Student(Student s) {  
        name = s.name;  
    }  
}
```